HD AXIAL SERVO CONTROL + DYNAMIC BRAKE

Stationary

TGPU

CPU/FPGA

LEDs:
DS10- Enc_CH_A (Grn)
DS11- Ax_Ready (Grn)
DS12- Ax_Brake (Grn)
DS13- Enc_CH_B (Grn)
DS14- HomeFlag (Grn)
DS15- Enc_CH_C (Grn)
DS16- Ax_CAN (Grn)
DS20- Ax_Cont_Cmd (Grn)
DS24- Ax_Scan_SW (Grn)
DS27- Ax_Cont_RDBK (Grn)

Gantry Tilt Frame

GANTRY POWER PAN
PDU 120 VAC
AC RTN

SSR

AXIAL HOLDING BRAKE (ENERGIZE TO RELEASE)

MOTOR SHAFT

AXIAL DRIVE

J29
Ax_Brake_Out
Ax_Brake_Out_Rt
Ax_Can_H
Ax_Can_L
+12V_Ax_Dr
HGND
Ax_Flt
Ax_Flt_Rtn
Ax_At_Spd*
Ax_Spd_Freq_Rtn
Ax_At_Freq*
Ax_At_Pos*
Ax_At_Pos_Rtn

J3 CAN SIGNALS

+12V_AX_DR

TB2
AX_FAULT_CONTACT
AX_SPEED_CONTACT
AX_FREQ_CONTACT
AX_POS_CONTACT

TB3
COMMAND SIGNALS
Ax_En_P
Ax_En_N
Start_Axial
Start_Stp_Com
Stop_Axial

Encoder Signals
Ax_ChA
Enc_A*
Enc_B
Enc_B*

J21
Ax_Enc_ChA
Ax_Enc_ChA_Rtn
Ax_Enc_ChB
Ax_Enc_ChB_Rtn
Ax_Enc_ChC
Ax_Enc_ChC_Rtn
VCC
LGND
Home_Flag
Home_Flag_Rtn

TB1
U V W
3 Phase HV PWM

AXIAL MOTOR
HD Motor Configuration

T1 T2 T3 T4 T5 T6 T7 T8 T9 P1 P2
3 Phase HV PWM
WIRED FOR **LV** MODE
NC
NC

GANTRY DRIVE BELT

ROTATING BASE

ENCODER GEAR

GANTRY ENCODER
(Generates 13 C-Pulses Per Gantry Revolution)

AX_ENC_CHA
AX_ENC_CHA_RTN
AX_ENC_CHB
AX_ENC_CHB_RTN
AX_ENC_CHC
AX_ENC_CHC_RTN
ENCODER_+5V
ENCODER_GND

HOME DETECT FLAG

GANTRY AZIMUTH DETECTOR
(Generates 1 Pulse per Gantry Revolution)

J1
ENCODER_+5V
ENCODER_GND
HOME_FLAG
HOME_FLAG_RTN

VCC TB3
FAULT

DYNAMIC BRAKE ASSEMBLY

Dynamic Brake Interlock
INTFC BD
120VAC PDU
120V_Rtn
FILTER BD
H1 H2 H3 H4
X1 X2 X3

INPUT POWER STAGE
TB1
R S T
DC+
DC-

REFER TO AXIAL DRIVE POWER CONTROL